

Supervised Learning

Definition:

Supervised learning is a type of machine learning where the model is trained on a labeled dataset — meaning the input data is paired with the correct output.



Example:

- **Email Spam Detection**

The model is trained with emails (input) and labels like "spam" or "not spam" (output).

- **House Price Prediction**

The model learns from features like square footage, location, etc., with known prices (target).



Characteristics:

- Uses **labeled data**.
- Goal: **Predict or classify** future data.
- Output is **known** during training.



Common Algorithms:

- Linear Regression
- Logistic Regression
- Decision Trees
- Support Vector Machines (SVM)
- Random Forest
- k-NN
- Neural Networks

Unsupervised Learning

Definition:

Unsupervised learning is a type of machine learning where the model is given input data **without any labeled outputs**, and it tries to find patterns, structures, or relationships in the data.



Example:

- **Customer Segmentation**
Group customers based on behavior without knowing the group labels beforehand.
- **Market Basket Analysis**
Find items that are frequently bought together.



Characteristics:

- Uses **unlabeled data**.
- Goal: **Find hidden patterns or groupings**.
- Output is **unknown** during training.



Common Algorithms:

- K-Means Clustering
- Hierarchical Clustering
- Principal Component Analysis (PCA)
- DBSCAN
- Association Rule Learning (e.g., Apriori)

Feature	Supervised Learning	Unsupervised Learning
Input Data	Labeled	Unlabeled
Output Known?	Yes	No
Goal	Prediction/Classification	Clustering/Pattern Discovery
Example Task	Email spam classification	Customer segmentation
Algorithms	SVM, Decision Tree, etc.	K-Means, PCA, etc.